Kerf Spike: checking the laser connection

Do this once on each of the four station PCs. It takes about 10 minutes. The spike never fires the laser in the normal test; it connects, reads the laser's settings, homes it and jogs it a few millimetres.

1. Before you start

- [ ] Plug the laser's USB cable into this PC and switch the laser on.
- [ ] **Close** LightBurn, LaserGRBL and any Creality software. Only one program can use the laser at a time.
- [ ] Clear the bed, or leave a scrap of material on it. The test **homes** the head and then moves it about 10 mm.
- [ ] Stay at the machine for the whole test and keep the stop button within reach.
- [ ] Copy the **kerf-spike** folder (or unzip kerf-spike.zip) onto the PC. No install and no administrator rights are needed.

2. Run the test

1. Open the kerf-spike folder and double-click **run_spike.bat**. If Windows says the file is blocked, ask IT to allow the folder.
2. When asked for the mode, type **a** (auto test) and press Enter.
3. Type the station name, for example **station1**.
4. Type the server address if you were given one, or press Enter to skip.
5. If it cannot choose the laser's port on its own it lists them: type the port name, for example **COM3**.
6. When it asks about the air assist, listen for the fan or pump and answer **y** or **n**.
7. Wait for **Done**. Note the line 'Send this ONE file back for diagnosis': it names a .zip file.

Press Ctrl+C at any time to stop. The laser is told to stop straight away and the results so far are still saved.

3. What each test checks, and what a pass looks like

#	Test	A pass means
1	List serial ports	The laser's USB port (a CH340 chip) is listed, for example COM3.
2	Open port, read GRBL banner	The laser answered with a line starting 'Grbl 1.1'.
3	Read settings ($$)	At least 10 settings came back. Note $30, $32, $130 and $131.
4	Status reports (?)	Five status lines in a row, e.g. <Idle|MPos:0,0,0|...>.
5	Home ($H)	The head homed and the laser returned to Idle. (A FAIL with error:9 means homing is switched off in the laser: $22=0.)
6	Jog 10 mm X/Y and back	The head moved 10 mm each way and the reported position changed and returned.
7	Air assist M8/M9	You heard the air assist switch on and off.
8	Feed hold, resume, soft reset	The laser paused, resumed and reset on command.
9	Camera probe	A camera picture was captured. SKIPPED is fine if there is no camera.
10	Outbound HTTPS to server	The PC can reach the Kerf Server. SKIPPED if you left the address blank.
11	Optional laser fire	Off unless you ran with --enable-fire. Leave it off for now.

4. Check the results

Open the **output** folder next to run_spike.bat, then the newest folder (the station name and the time). Open **report.txt**.

[ ] Tests 1 to 8 should say **PASS**. Tests 9 to 11 may say SKIPPED.

[ ] Write down the values on the record sheet below, from **settings.txt**: **$30** (maximum laser power value), **$32** (must be 1: laser mode on), **$130** and **$131** (bed width and height in mm).

[ ] Did the head really move in tests 5 and 6? Did you hear the air assist in test 7?

[ ] Anything marked FAIL: do not worry, just send everything back (step 6). The logs say why.

5. If something goes wrong

What you see	What it usually means	What to try
'No serial ports found'	The USB driver is missing, or the cable is a charge-only cable.	Try another USB cable and port. Tell Mr Salih if the port still does not appear (the CH340 driver may be needed).
'Could not open COM3 ... access is denied'	Another program has the laser's port.	Close LightBurn, LaserGRBL, Creality software and any serial monitor, unplug and replug the cable, then run again.
'No Grbl 1.1x banner received'	Wrong port, wrong speed, or the laser is off.	Switch the laser on, check the port name, run again. Send the logs back.
Test 5 FAIL, error:9	Homing is disabled in the laser ($22=0).	Not a fault in the spike. Note it on the sheet; later tests carry on.
Test 5 FAIL, ALARM	A limit switch or a hard fault.	Check nothing is blocking the head, switch the laser off and on, run again.
Window closes or freezes	A driver or camera problem.	Wait 3 minutes (it records where it is stuck), close the window, then send the logs back.

6. Send the results back

Always send them back, even if every test passed. The one .zip file in the output folder holds everything. Send **that one file** (email, shared drive or USB stick) and say which station it is.

Inside the zip, for whoever diagnoses it:

File	What it holds
session.log	One timeline with times: every command sent to the laser, every reply, what was shown on screen, what was typed, warnings (timeouts, error replies) and any error with its full detail.
report.txt / report.json	Pass, fail or skipped for each test, with details. Saved after every test, so a crash still leaves it.
settings.txt	The laser's full settings ($$).
env.txt	The PC, Python, serial driver, USB devices and camera information.
serial.log	The same serial traffic in a machine-readable form, with the raw bytes.
crash.log	Details if a driver crashed or a test got stuck.

7. Record sheet (one per station)

Station		Date	
GRBL banner		Name of the zip sent	
$30 (laser power max)		$32 (laser mode, want 1)	
$130 (bed width, mm)		$131 (bed height, mm)	
Home worked? (test 5)		Jog worked? (test 6)	
Air assist heard? (test 7)		Camera found? (test 9)	
Notes			

Run the spike on all four station PCs before the real Cutter is built: its settings come from these results.